Demonstration of proposed features

Date	12 July 2026
Team id	XXXXX
Project name	Opti crop: smart Agricultural production optimization Engine
Maximum marks	1 mark

Demonstration of proposed features:

S.No	Feature Name	Description	Status (Implemented / Partial / Pending)	Demonstrated (Yes / No)	Remarks
1	User Input Module	Accept soil nutrient and environmental parameters from the user.	Implemented	Yes	Working successfully
2	Data Preprocessing	Clean and preprocess the agricultural dataset before training.	Implemented	Yes	Successfully tested
3	Machine Learning Model	Train the Random Forest model for crop recommendation.	Implemented	Yes	Accurate predictions generated
4	Crop Recommendation	Predict the most suitable crop based on user inputs.	Implemented	Yes	Accurate crop recommendation displayed
5	Flask Web Application	Provide a userfriendly web interface for crop prediction.	Implemented	Yes	Responsive and easy to use
6	Model Integration	Integrate the trained Random Forest model with the Flask application.	Implemented	Yes	Real-time prediction achieved

Feature implementation summary:

S.No	Challenge Faced	Resolution / Action Taken
1	Difficulty coordinating team schedules.	Conducted meetings at mutually convenient times and shared updates through WhatsApp.
2	Delays in completing assigned development tasks.	Redistributed tasks among team members and monitored progress regularly.
3	Issues during machine learning model integration with the Flask application.	Collaboratively debugged the application, tested the integration, and resolved compatibility issues before deployment.